

Activity 9 Pre-Lab — EAS Bromination of Aniline / N-Phenylacetamide

Organic Lab · electrophilic aromatic substitution

[Your name] & [partner]
[MM/DD/YY]

■ **Black = write this in your notebook** ■ **Blue = context, don't copy it** Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Study **electrophilic aromatic substitution (EAS)**: first **protect** aniline's amine as an amide (N-phenylacetamide), then **brominate** the ring. This shows how an –NHR group is a strong **o/p-activator** and how protection tempers its reactivity so you get clean **para** product.

👉 **DRAW** — the two steps

1) aniline + acetic anhydride → **N-phenylacetamide** (acetanilide) + acetic acid

2) N-phenylacetamide + “Br⁺” → **4-bromo-N-phenylacetamide** (para) Br⁺ made in situ from NaBr + NaOCl (bleach) in acetic acid; amide is an o/p-director → para major

Chemical Reagents & Safety Table

CONTEXT ▶ Required (8): aniline, acetic anhydride, sodium hypochlorite, 95% ethanol, glacial acetic acid, sodium bromide, sodium thiosulfate, sodium hydroxide. Typical SDS values — cross-check the SDS tile. **Aniline is quite toxic** — treat carefully.

Chemical	Role	MW	Amount	Physical properties	Hazards (GHS)
Aniline	Starting material (limiting)	93.13	~1 mL (0.5 mL x2, ~0.011 mol)	Colorless–brown oily liquid; bp 184 °C; d 1.02 g/mL	H301+H311+H331 toxic (swallow/skin/inhale) · H318 serious eye damage · H317 skin sensitizer · H341 susp. mutagen · H351 susp. carcinogen · H372 · H400
Acetic anhydride	Acetylating (protecting) agent	102.09	2 mL	Colorless liquid; bp 139 °C; d 1.08 g/mL	H226 flammable · H302 · H314 severe skin burns & eye damage · H332 harmful if inhaled
Sodium hypochlorite (bleach)	Oxidant → generates Br ⁺	74.44	18 mL	Pale greenish aqueous liquid	H314 severe skin burns & eye damage · H400 aquatic · EUH031 toxic gas w/ acid
Sodium bromide	Bromine source	102.89	~2.8 g	White crystalline solid	Low hazard · H319 eye irritation
Glacial acetic acid	Solvent / acid medium	60.05	10 mL	Colorless liquid; bp 118 °C; d 1.05 g/mL	H226 flammable · H314 severe skin burns & eye damage
95% Ethanol	Recrystallization solvent	46.07	10 mL	Colorless liquid; bp 78 °C; d 0.79 g/mL	H225 highly flammable · H319 eye irritation
Sodium thiosulfate	Quenches excess oxidant	158.11	as directed	White solid (aq. soln)	No significant GHS hazards
Sodium hydroxide (0.5 M)	Dissolves/works up amide	40.00	3 × 10 mL	Colorless aqueous solution	H290 corrosive to metals · H314 severe skin burns & eye damage

GRAB FIRST: 50-mL round-bottom + stir bar · stir plate + clamp · ice-water bath · 2× 1-mL syringes (aniline) + 25-mL Erlenmeyer + 400-mL beaker · latex gloves (aniline) then nitrile · steam bath · 125-mL Erlenmeyer · Büchner funnel + filter flask + vacuum · graduated cylinders.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Protect: aniline → acetanilide

1. Instructor adds **2 mL acetic anhydride** to a clamped **50-mL RBF** + stir bar; stopper; lower into an **ice bath** on a stir plate.
2. At the aniline hood (**latex gloves**), draw **0.5 mL aniline into each of 2 syringes** (record exact volume for theoretical yield). **Aniline is toxic + absorbs through skin — don't overdraw the syringe.**
3. Add the aniline **dropwise over 5 min** with stirring → mixture solidifies (acetanilide); break up lumps with a spatula. Clean syringes with acetone into aniline waste; switch to **nitrile gloves**.
4. Add **10 mL 0.5 M NaOH**; warm on steam bath ~1 min to loosen solid; pour into a 125-mL Erlenmeyer; rinse RBF with **2 more × 10 mL NaOH** into the same flask.
5. Cool to RT then in ice ~10 min; collect the colorless **N-phenylacetamide** by **vacuum filtration**; save an IR-size sample + take IR.

Part 2 — Brominate

6. Dissolve the acetanilide in **10 mL glacial acetic acid + 10 mL 95% ethanol**; add ~2.8 g **NaBr**, stir to dissolve.
7. Measure **18 mL bleach (NaOCl)** and add to the reaction; stir. **Bleach + NaBr in acid makes the Br⁺ electrophile in situ. Never mix bleach with straight acid elsewhere — toxic gas.**
8. Quench excess oxidant with **sodium thiosulfate**, precipitate, **vacuum filter**, and **recrystallize from 95% ethanol**; weigh → % yield; mp / IR / NMR (para pattern).

✍️ WRITE

aniline is limiting → theoretical acetanilide → theoretical 4-bromoacetanilide; % yield each step.

Experiment Notes (in lab)

Exact aniline volume: _____ mL

Part 1 — solidifying / color:

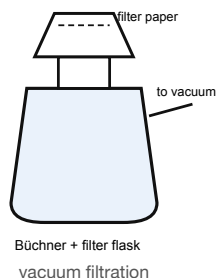
Acetanilide mass + IR notes:

Part 2 — on adding bleach (color/heat):

Product mass / mp: _____

% yield · para evidence (NMR):

Setup — draw this



CONTEXT (don't copy) ▶

Why protect the amine first? free aniline is SO activated it over-brominates (tri-bromo) and its lone pair also reacts with the oxidant. Acetylating to the amide dials the activation down to a controllable **o/p-director**, so you get mono-**para** product.

Where does Br⁺ come from? NaOCl (bleach) oxidizes Br⁻ (from NaBr) in acetic acid to an electrophilic bromine species — greener than using liquid Br₂.

Clean-up & Conclusions

- Aniline rinses → labeled aniline waste; latex gloves → solid waste; filtrates → aqueous waste; organic solvents → organic waste; keep bleach away from acids; bench task.
- **Conclusions** (~4 sentences **after lab**): % yields; that the **-NHC(O)CH₃** group directed bromination **para** (NMR: two doublets, para pattern); why protecting the amine was necessary; a loss source.